

Final Report

Levitating Rose Illusion

ENG ME 461 - Senior Design II
Section A1
Project #205

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Introduction and Background

Abstract

This project involves the design and development of an electromechanical enchanted rose prop for a live production of *Beauty & the Beast*. Mounted on an L-shaped arm within a display enclosure to create the illusion of levitation, the rose features twelve individually actuated petals that are ejected by the solenoids triggered through a DMX control board, which is the industry standard for live stage effects. As the play progresses, petals are dropped on cue, serving as a visible countdown to the Beast's fate. The core engineering challenges include reliable solenoid actuation, DMX integration, and repeatable petal ejection under live performance conditions.

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Executive Summary

The Levitating Rose Project is the development of a stage-ready theatrical prop for productions of *Beauty and the Beast*, designed to create a believable levitating rose with a controlled petal release mechanism and integrated lighting effects. The system combines mechanical, electrical, and structural subsystems into a cohesive product that prioritizes reliability, repeatability, and ease of operation in live theater environments.

Acknowledgements

Our team would like to thank and acknowledge the many people who have aided in the design of testing of our product, including but not limited to: Evan Northrup, our customer who has aided and supported us with his guidance throughout every step of both the design and testing process of our product. Eric Hazen, the Electronics Advisor at Boston University who has helped us design and implement our complicated electronics system and integrate it into existing DMX systems. Ben Sigda, the Lighting and Sound Shop and Production Manager at the Joan & Edgar Booth Theatre in Boston University, who generously let us test and troubleshoot our final product in a professional theatre environment using the proper equipment. Katie Kelso and Bianca Sparr for guidance on all of the woodworking machinery at the Singh Imagineering Laboratory.

As for the design elements and their origin, the idea of using push pull solenoids for the petal release and using UV paint for the magical glow came from our team member Anna Tretiakova, the enclosure and its victorian era design came from our team member Danna Martinez, and the design for the bracket holder and petal hooks came from Nicolas Martinez. It is worth noting that our customer, Evan, aided in the creation of all of these design elements.

Motivation

This project was motivated by the unique opportunity to apply mechanical engineering principles and design to a real world theatrical application. *Beauty & the Beast* is a widely recognized franchise with an established prop, the enchanted rose, that has a specific and well defined role in the story. Recreating it as a functional, user controlled electromechanical system presented a meaningful engineering challenge while also producing a tangible deliverable for a live production. The team selected this project because it bridges engineering and performing

arts, a rare and unlikely combination of aspects in engineering fields of study. The prospect of observing the system perform in a live production setting offered a unique opportunity for real-world testing and validation not typically available in a traditional capstone project.

Problem Statement

Live theatrical productions of *Beauty & the Beast* require a functional enchanted rose prop that can reliably shed individual petals throughout a performance. Existing solutions typically rely on manual operation that require lengthy setup times and lack precision and repeatability. This project addresses the need for a fully integrated DMX controlled electromechanical system capable of actuating twelve individual petals, while maintaining the aesthetic quality of a theatre production.

Research, Benchmarking and Competitive Assessment

The primary resource for hardware research was the project's client, Evan Northrup, who provided insight into how existing enchanted rose props are currently implemented in theatre productions. Existing solutions rely on a manual string-pull system, in which stagehands hold individual strings attached to rose petals and release them at the appropriate moment during the performance. While functional, this approach has notable limitations: it requires dedicated personnel during the production, is prone to tangling as petals fall, and results in lengthy reset times between performances.

Two alternative petal release mechanisms were evaluated before arriving at a final design. The first was a motor-driven string system, in which motors would automate the existing string-pull approach by unclipping petals on cue. While this addressed the need for dedicated personnel, it did not resolve the issues of reset time or string tangling and was therefore eliminated. The second option considered was electromagnetic retention, in which petals would be held in place magnetically and released by cutting power. This was rejected due to the significant size and power requirements of electromagnets capable of reliably holding and releasing the petals. Based on this assessment, solenoid-driven petal ejection was selected as the optimal solution, offering a compact, low-power, and repeatable mechanism for releasing individual petals on cue. These ideas for petal release were all screened by our customer, Evan, who served as a technical expert in our design process.

Customer Requirements

Customer

The customer of this project is Evan Northrup, a professional magician and illusion designer. He has a background in the performing arts, having spent years in musical theater, dance, and studying magic formally. He works in close collaboration with theater and arts organizations designing illusions for stage productions. He also offers theater consulting and trade-show design services. His creative philosophy emphasizes intention, curiosity, and storytelling, aiming to fuse psychology and performance to create magic experiences that linger for audiences.

Product Requirements

- Repeatability and reliability: The rose must levitate and release petals consistently across multiple performances without mechanical failure.
- Visual impact and believability: The illusion should be compelling, with the rose appearing to float naturally and petals falling gracefully.
- Ease of operation: Theater personnel should be able to operate the system with minimal training and no specialized technical knowledge.

Engineering Specifications

- Petal release mechanism: There should be a minimum of 7 and a maximum of 12 petals. The system needs to be reset in under 5 minutes and be done by people with average hand dexterity.
- Lighting system: If UV reactive paint is used, the paint and lights need to be in the same wavelength (365–370 nm). All lighting needs to be powered from wall outlets to avoid battery dependency, and operate with 12 volts and 1 amp current.
- Enclosure: A stable stand capable of maintaining the rose upright while withstanding vibrations from stage movement and handling. The enclosure should be no smaller than a 1 ft x 1 ft x 1 ft volume and no bigger than a 2.5 ft x 2.5 ft x 2.5 ft volume.

Engineering Specifications

The following final designs were heavily influenced by the input and guidance of the project's client, Evan Northrup. The final petal release design centers on a custom solenoid enclosure that houses twelve solenoids, each seated in a dedicated slot within the enclosure body. Each slot features an opening positioned directly above the solenoid push pin, through which a petal hook is inserted. Petal hooks are small 3D printed connectors that interface between the solenoid and the rose petal. On one end, a linen fabric rose petal is affixed via hot adhesive, while the other end seats into the solenoid slot. When actuated, the solenoid push pin ejects the petal hook, releasing the petal. Dimensioned drawings for the solenoid enclosure and petal hook components are provided in the Appendix.

The stem holder is a 3D printed L-shaped bracket mounted approximately one foot above the floor of the enclosure. The bracket is bolted directly to the rear interior wall of the enclosure, positioning the rose at an elevated height while concealing the support structure against the back wall to create the visual illusion of levitation. The rose stem is secured to the bracket via a circular clamping bracket that wraps around and grips the stem. Dimensioned drawings for the L-bracket, clamping bracket, and full assembly are provided in the Appendix.

As part of our hand off procedure, we have created a full and inclusive manual for both our customer and the theatre group working with our product, should there ever be one. This manual serves a full users manual with a high level overview of the systems of the rose, along with detailed instructions with visual aid for both setup and reset procedures. This manual will be given to our customer along with any and all files necessary for the creation of the product, including all STL files needed for 3D printing any parts that may get lost or broken.

Design Description

System Overview



Figure 1: The fully completed levitating rose project

The Levitating Rose system is an integrated theatrical prop designed to simulate a floating rose with a controlled, sequential petal release mechanism and synchronized lighting effects. The primary subsystems include: (1) The rose bud and petal release mechanism, which houses push-pull solenoids arranged in a layered, numbered configuration to enable controlled petal actuation. (2) The mechanical stem support and bracket system, which secures the rose while preserving the illusion of levitation. (3) The enclosure assembly, constructed from birch plywood and frosted acrylic panels, which provides structural stability, spatial integration, and concealment of wiring and electronic components. (4) DMX controlled electronics, consisting of a RMAX485 module, Arduino, transistor networks, mosfets, UV and LED lights. These subsystems are physically and electrically integrated to ensure repeatable petal release, complete DMX compatibility, and structural stability under stage handling and movement. The enclosure provides structural support while also improving the overall appearance. It holds the mechanical components in place, routes the electrical connections, and conceals internal elements within the available space.

Subsystems

Bud Holder

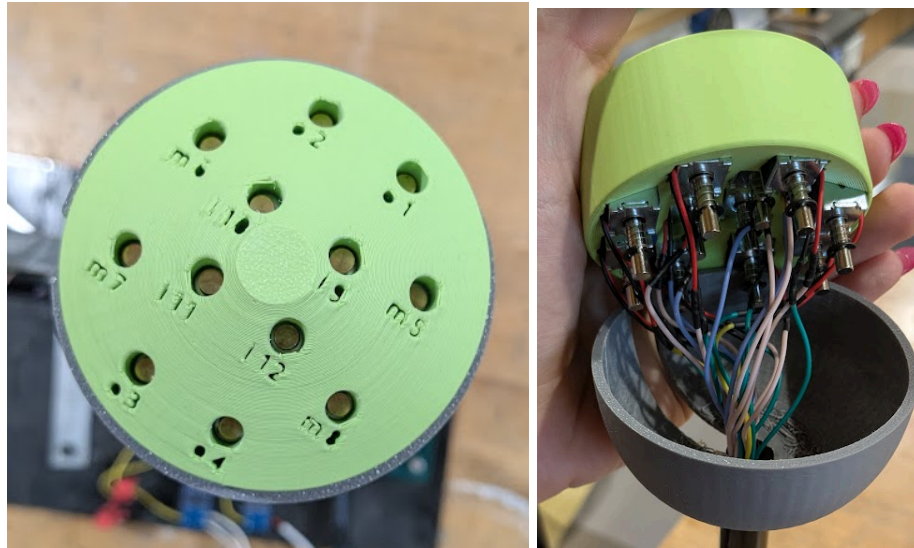


Figure 2: The bud, both viewed from above and positioned to show the solenoids

The bud/solenoid holder features a conical shape and compact form factor, making it less visually prominent and contributing to the illusion of the product being a natural rose, while also ensuring consistent petal release. The holder contains 12 solenoid inserts numbered 1 through 12, organized across three layers: outer, middle, and inner. These layers are physically labeled on the holder with their respective initials (O, M, I). The outer layer houses solenoids 1–4, the middle layer houses solenoids 5–8, and the inner layer houses solenoids 9–12. The solenoid numbers increase counter-clockwise within each layer. This labeling system - combining a letter to denote the layer and a number to denote the solenoid - is physically embedded into the holder for easy identification.

Petal Hooks

To connect individual rose petals to each solenoid, custom Petal Hook components were designed and glued to petals cut from a plastic rose. Each hook is designed to be sturdy and resistant to accidental breakage during handling. Two petals are attached per hook, resulting in a total of 24 petals across all 12 solenoids. This arrangement gives the rose a fuller, more convincing appearance that closely resembles a natural bloom.

Stem Holder

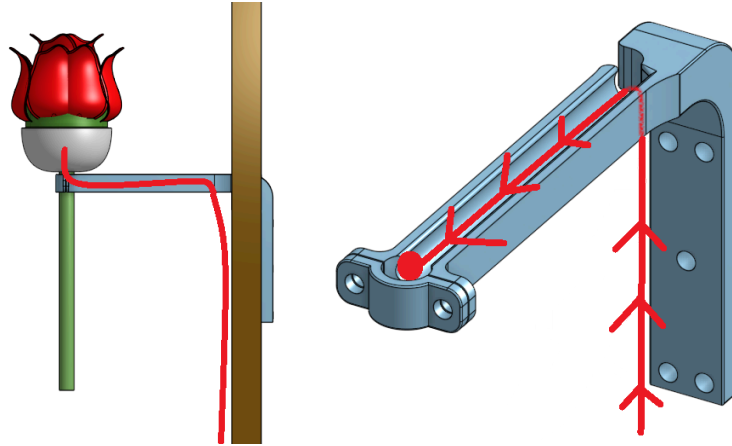


Figure 3: CAD images of the stem holder, from multiple angles. Cable location indicated in red.

To hold the rose itself and the enclosure, an L shaped holder was designed and 3D printed. The stem holder is bolted to the interior of the back wall of the enclosure using M6 bolts (depicted in Figure 1.2.2), and a circular clamp secured with M6 bolts holds the rose stem in place. This configuration creates the visual effect of the rose floating or levitating in mid-air. To maintain the illusion, the holder is concealed from the audience using practical visual elements such as greenery and leaves, in accordance with the customer's specifications. The holder also features a circular cutout to allow wiring to run cleanly from the back wall of the enclosure to the rose bud.

Enclosure

The enclosure subsystem provides structural support, spatial integration, and visual concealment for the levitating rose system. It houses the electronics, supports the mechanical mounting of the rose, and enhances the illusion through controlled visibility.

The finalized dimensions are 20 in $\times$ 20 in $\times$ 30 in, remaining within the specified 1 ft $\times$ 1 ft $\times$ 1 ft to 2.5 ft $\times$ 2.5 ft $\times$ 2.5 ft constraints. The increased height accommodates the full rose assembly and required vertical clearance, while the 20-inch width and depth provide adequate space for petal release and internal wiring without compromising stability. The structure consists of a birch plywood base and top connected by vertical frame members, providing rigidity under stage handling. Electronics are housed in the base to lower the center of gravity and improve stability. The walls are constructed from 1/4-inch frosted acrylic panels, which diffuse light and

obscure internal components while maintaining visibility of the rose. The enclosure therefore functions as both a load-bearing structure and an illusion-enhancing element, integrating mechanical and electrical subsystems into a stable, stage-ready design.

The resulting design prioritizes mechanical clearance and visual presentation while still satisfying the functional internal volume requirements of the system. Detailed CAD views of the enclosure assembly are provided in Appendix Figures A.4 and A.5.

Electronics

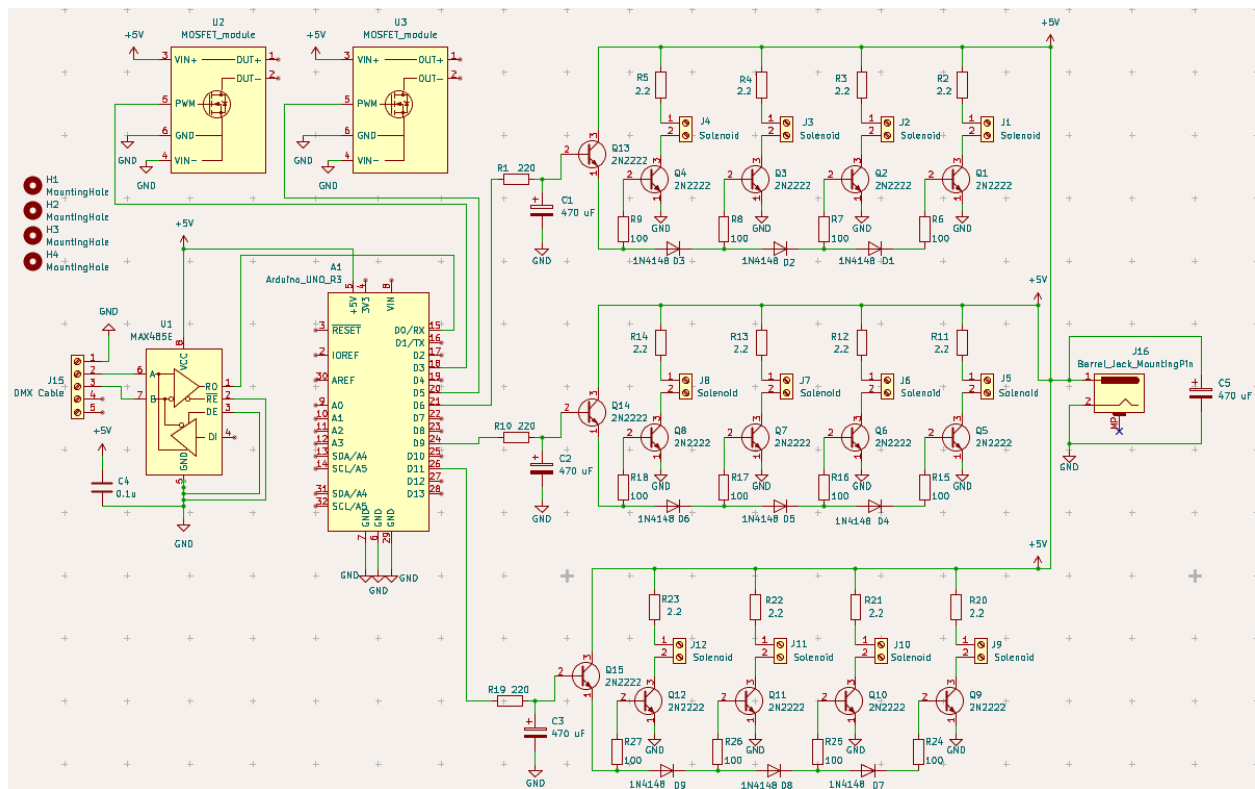


Figure 4: Complete Electronics Schematic

The electronics subsystem allows the DMX to fully control all the individual parts of the rose. The schematic illustrated in Figure A.6 depicts a DMX-controlled 5 V switching PCB, built around a MAX485 RS-485 transceiver and an Arduino. The MAX485 converts DMX512 differential signals to TTL serial for the Arduino, which decodes DMX channels and drives multiple output channels via transistor networks and MOSFET driver stages. The transistor networks sequentially activate four push-pull solenoids one at a time as the Arduino outputs an increasing control value. There are three identical transistor networks, giving a total of 12

solenoids, meaning the rose will have 12 release points, each actuating a hook that holds two cosmetic petals for a total of 24 petals released during the blooming sequence. The UV lights cause the UV-reactive paint on the petals to fluoresce, producing a magical glowing effect, while the LEDs illuminate the entire rose to enhance its visibility on stage. The PCB takes a single +5 V supply (barrel jack) and needs a minimum of 10 amps to operate.

Design Decisions

The enclosure design evolved through several iterations to balance structural integrity, visual appearance, and functionality. One key decision was the selection of a rigid frame structure using birch plywood instead of lightweight materials such as foam board or thin plastics. While lighter materials would reduce weight, they lacked the stiffness required to withstand handling during transport and stage setup.

Another major decision involved the use of frosted acrylic panels rather than clear or opaque materials. Clear panels were initially considered but rejected because they exposed internal wiring and components, reducing the illusion effect. Fully opaque panels were also rejected because they blocked visibility of the rose. Frosted acrylic provided a compromise by diffusing light and partially concealing internal components while maintaining visual focus on the rose.

The enclosure dimensions were also iterated. A smaller design was initially considered but resulted in insufficient clearance for petal release and internal wiring. Increasing the height to 30 inches allowed for proper spacing and improved the visual proportions of the levitating rose.

Finally, the placement of electronics in the base was a deliberate decision to improve stability by lowering the center of gravity. This reduced the risk of tipping during handling and ensured safer operation in a live performance environment.

Design Evaluations

The enclosure was evaluated based on stability, structural integrity, and functionality during operation. Stability testing was performed by applying external disturbances

representative of stage handling and observing any tipping or excessive vibration. The enclosure remained stable due to its wide base and low center of gravity.

Structurally, the frame showed no visible deformation under the weight of the mounted rose and internal components. The birch plywood structure provided sufficient rigidity, and the connections between components remained secure during repeated handling.

Functionally, the enclosure successfully housed all subsystems, including electronics and wiring, without interfering with the petal release mechanism. The internal layout allowed for clean cable routing and easy access for maintenance.

From a visual standpoint, the frosted acrylic panels effectively diffused light and reduced visibility of internal components, enhancing the illusion of levitation. Overall, the enclosure met all design requirements related to stability, functionality, and aesthetics.

Product Cost Assessment and Team Budget Report

To build one additional unit of the levitating rose, it would cost approximately \$615.23. A majority of this cost goes to the materials required to build the enclosure and the electronics required to operate the rose. The materials for the enclosure include $\frac{1}{2}$ and $\frac{3}{4}$ in wood orders and acrylic plexiglass. It is worth noting that our vendor for these supplies was limited to Amazon only, and it is very likely that the wood and acrylic could be sourced from a big box store for a lesser price. The electronic components for our build, like the solenoids and DMX cables, were similarly sourced from Amazon, with the exception of the PCB board.

Overall our team was significantly over budget, with an initial budget of \$300, later being increased to \$400 with the addition of our final team member, compared to our final cost of the project being \$723.91. This puts us 81% over budget due to material and electronics cost. However, given that this project deals with not just engineering elements, but also artistic visuals and effects up to standard for professional theatre use, the initial goal of \$400 was a hard target to hit. An itemized list of everything our team ordered can be found in the appendix.

It is also worth noting that the cost of our project is slightly more than the cost of producing a new unit due to a mistake we made in our first PCB order which resulted in us having to order a second PCB.

Conclusions

The Levitating Rose project successfully achieved its goal of creating a stage-ready, electromechanical prop capable of producing a convincing levitation illusion with controlled petal release. By integrating mechanical, electrical, and structural subsystems, the team developed a system that is both functional and visually compelling for live theatrical use.

The final design met the main customer requirements, including reliable and repeatable petal actuation, ease of operation, and strong visual impact. The use of DMX control allows for precise timing during performances, while the solenoid-based mechanism provides consistent and independent petal release. Additionally, the enclosure subsystem contributed to both the stability of the system and the enhancement of the illusion by concealing internal components while maintaining visibility of the rose.

Testing and evaluation demonstrated that the system performs reliably under expected operating conditions, with stable structural behavior and effective integration of all components. While there are opportunities for further refinement, such as reducing system complexity and improving ease of reset, the current design meets performance expectations and is suitable for real-world application.

Overall, the project meets and in some aspects exceeds the customer's expectations by delivering a unique and functional solution that combines engineering design with theatrical creativity.

Lessons Learned

Throughout this project, several key lessons were learned. First, early integration of mechanical and electrical systems is critical to avoid redesign later. Second, custom PCB development requires careful planning, testing, and iteration to prevent costly errors. Third, transistor networks are effective but must be thoroughly validated for consistent performance. Fourth, prototyping early helps identify mechanical weaknesses. Fifth, clear wire management is essential for reliability and maintenance. Sixth, designing for ease of use improves real-world functionality. Seventh, testing in the actual environment is invaluable. Finally, budgeting must account for unexpected costs and redesigns.

Appendix

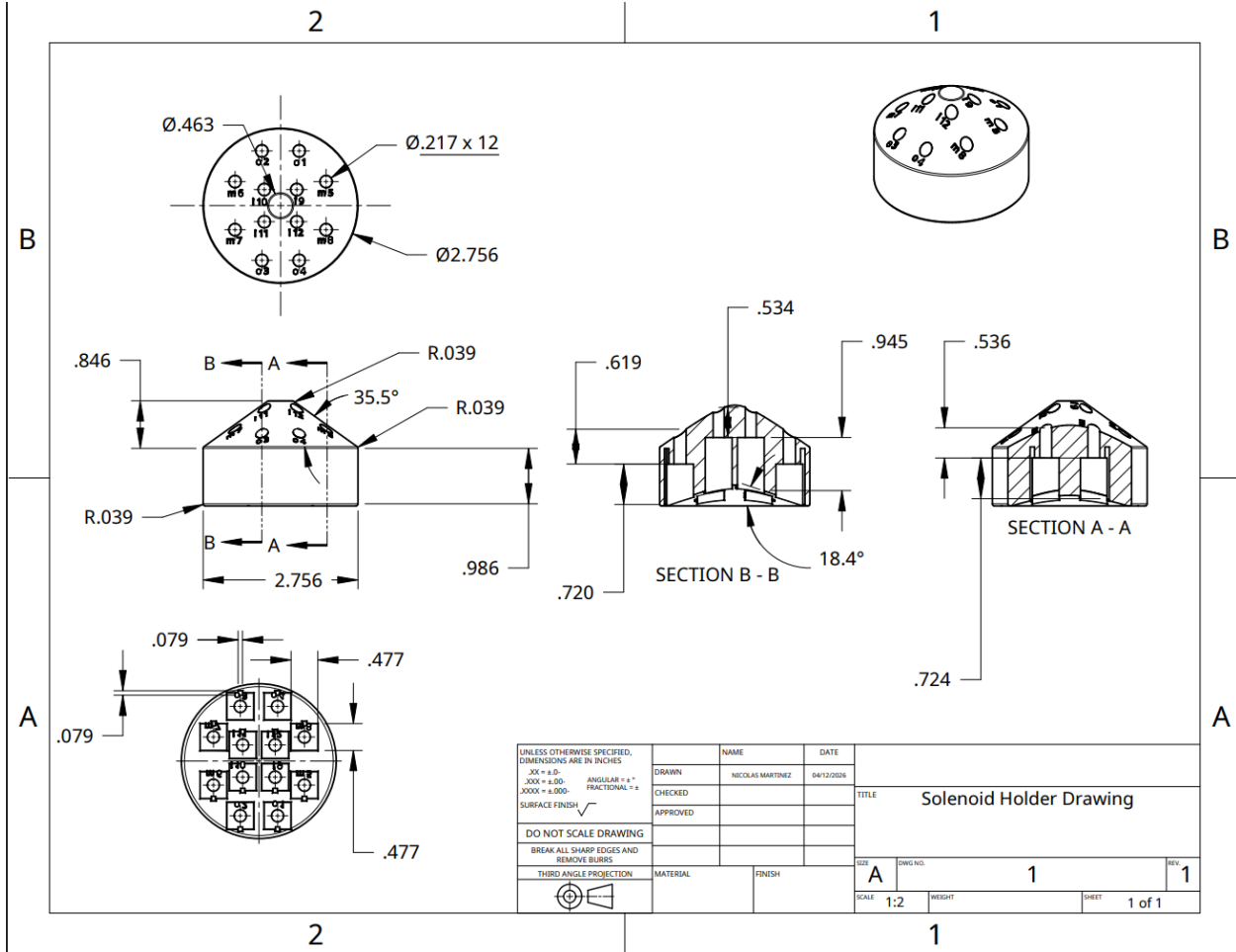


Figure A1: Drawing of the Solenoid Bud Holder

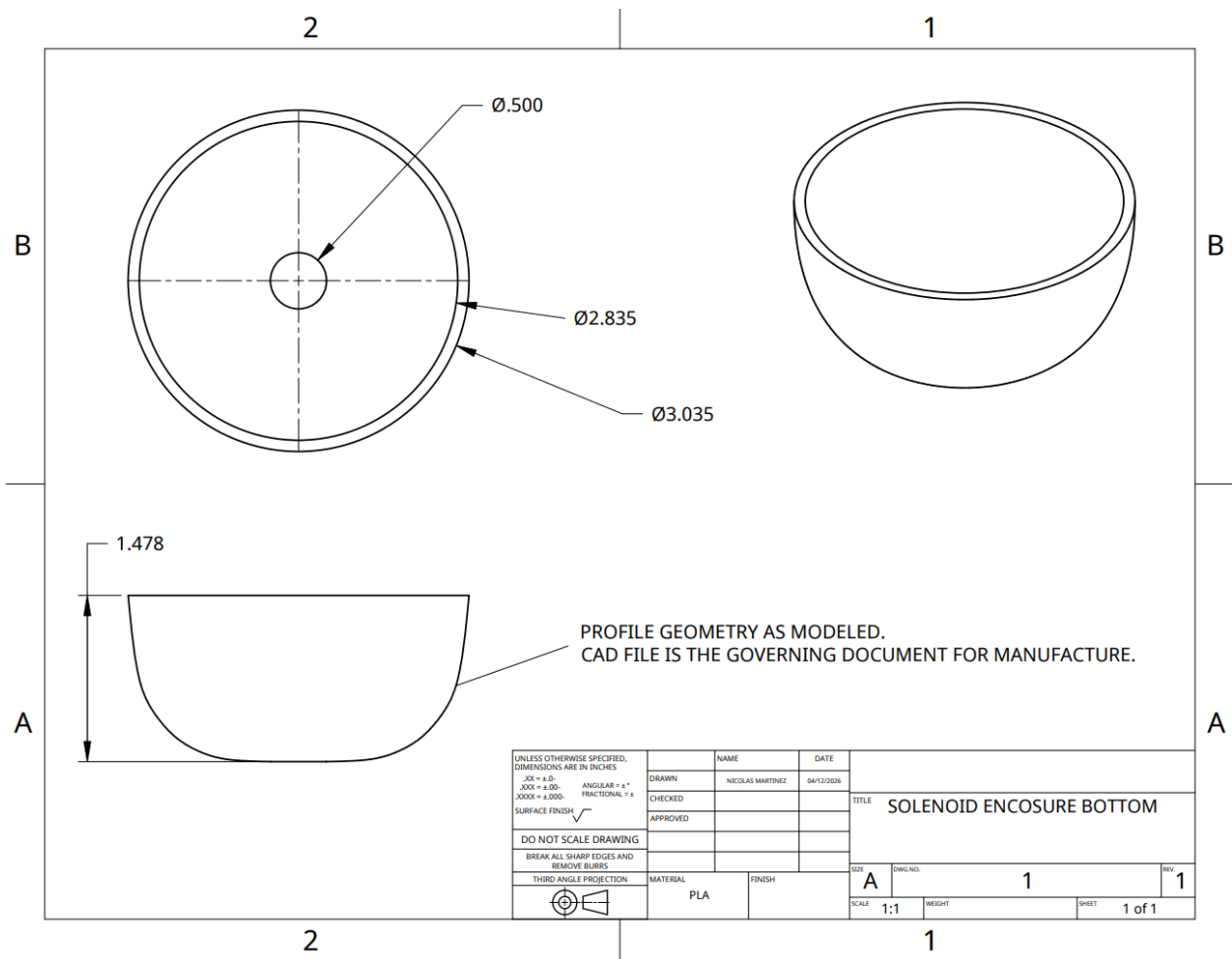


Figure A2: Drawing of the Bottom Solenoid Holder

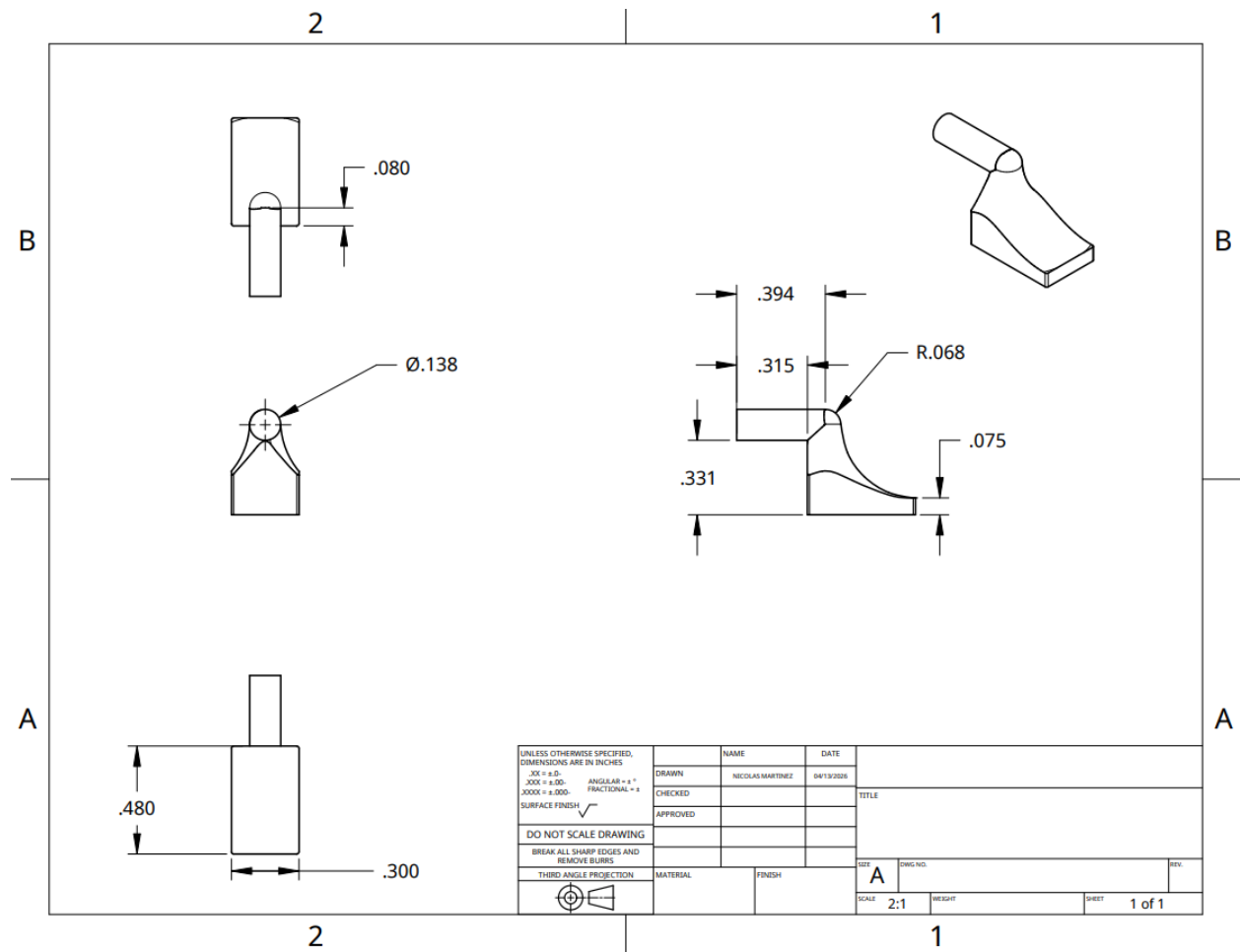


Figure A3: Drawing of a Single Petal Hook

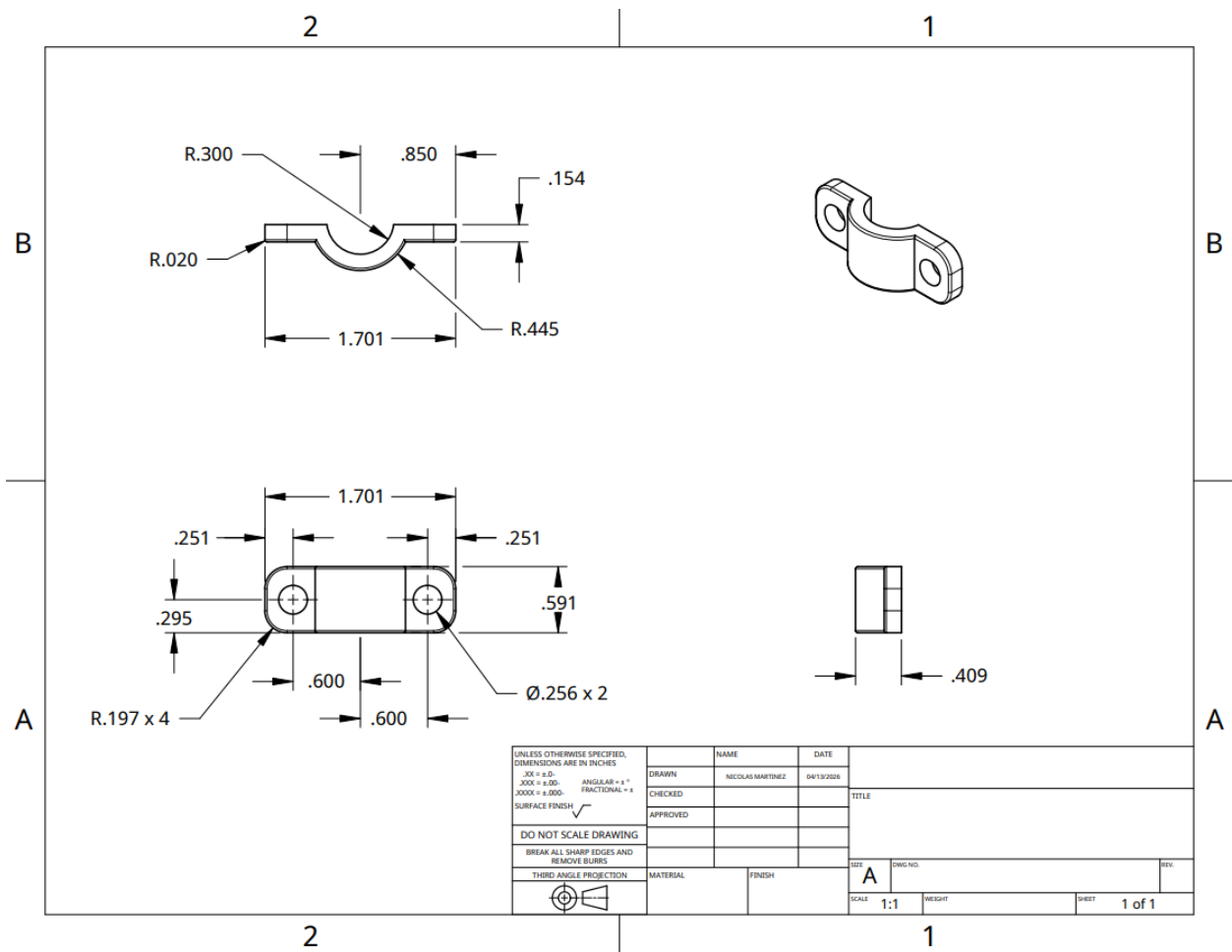


Figure A4: Drawing of the L-Bracket Part 1

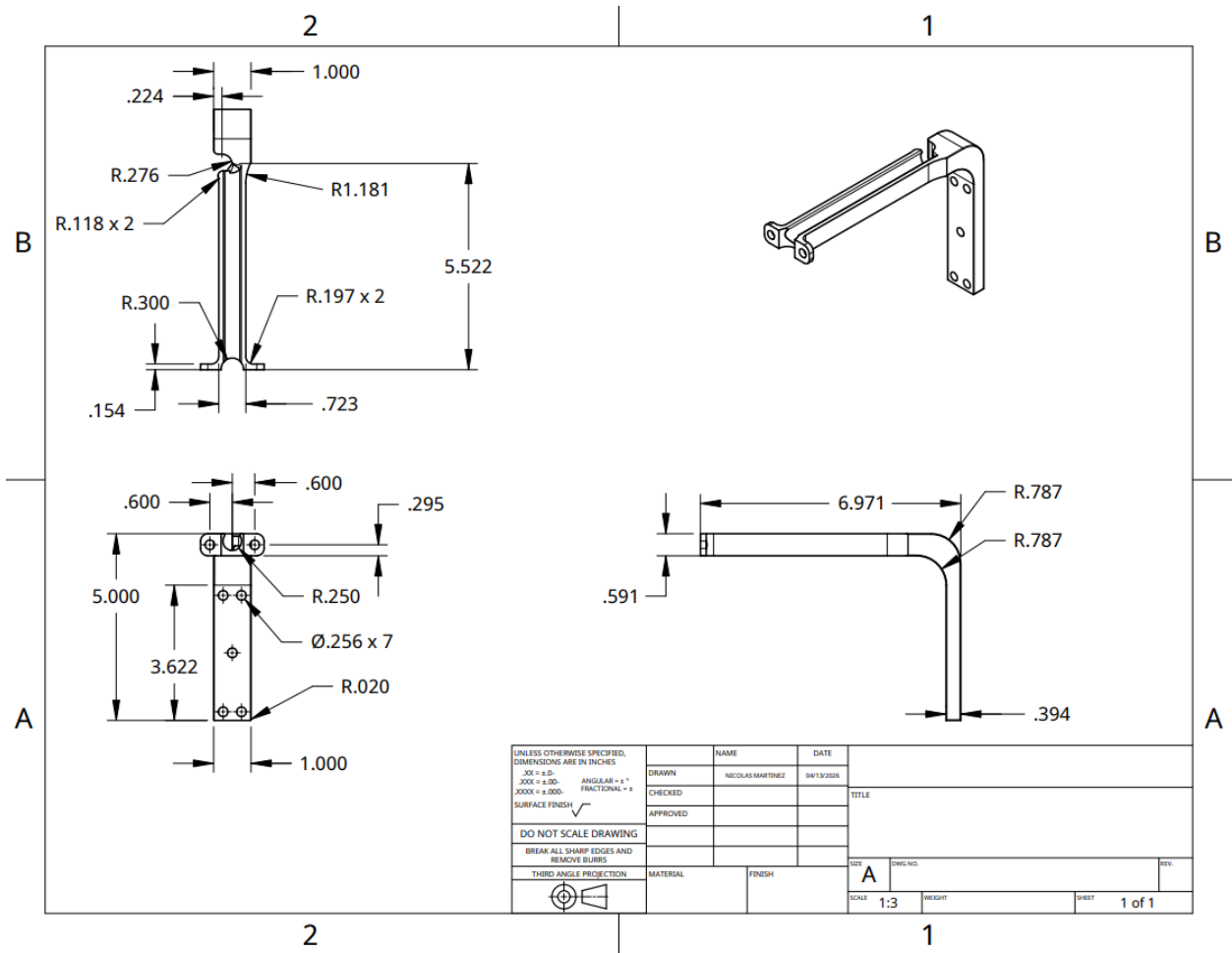


Figure A5: Drawing of the L-Bracket Part 2

Name	Model No.	Description	Vendor	QTY	Total Price
Silk Roses	B098D6DRWF	Artificial Roses for use in Prototyping (Pack of 9)	Dilatata (https://www.amazon.com/dp/B098D6DRWF?ref=fed_asin_title&th=1)	1	\$16.99
Glass Paint	72.604	Paint that creates a "frosted" effect on acrylic for enclosure	Vallejo (https://www.amazon.com/dp/B0BL3M3Z24?ref=fed_asin_title&th=1)	1	\$6.32
UV Paint	B00BD4UZ9I, B00BD4YIF0, B00BCTXWHG	Paint that only lights up when in contact with UV light. For use in enclosures.	GLO Effex (https://www.amazon.com/dp/B00BD4UZ9I?ref=fed_asin_title&th=1)	3	\$41.98
UV LEDs	UVA3W365NM	UV LEDs to light up UV paint. For use in prototyping	Cleiscry (https://www.amazon.com/dp/B0C5SW68P3?ref=fed_asin_title&th=1)	1	\$10.99
Push-Pull Solenoids	B0BRYJS7HD	Push-Pull Solenoids used for the Petal Release Mechanism	TEHAUX (https://www.amazon.com/dp/B0BRYJS7HD?ref=fed_asin_title&th=1)	1	\$29.68
3 Pin XLR Cable	PHX3M	Cable used to connect Rose electronics to DMX Boards	LCMLA (https://www.amazon.com/dp/B0G6Z5JTW8?ref=fed_asin_title&th=1)	1	\$12.59
USB-RS485-WE-1800-BT Cable	USB-RS485-WE-1800-BT	Cable used in electronics	FTDI (https://www.amazon.com/dp/B01HI9CMT2?ref=fed_asin_title)	1	\$26.99
½ Inch Wood	B0CQXTR7DD	Wood used for the enclosure	Woodpeckers (https://www.amazon.com/dp/B0CQXTR7DD?ref=fed_asin_title&th=1)	2	\$111.56
3 Pin XLR Male to Female Connectors	RD-XLR-MF	XLR Male to Female connectors (2 pack)	RonDexy (https://www.amazon.com/dp/B0F8MX6QQ8?ref=fed_asin_title&th=1)	1	\$12.98

¾ Inch Wood	B0BXBK5JTQ	Wood used for the enclosure	MakerStock (https://www.amazon.com/dp/B0BXBK5JTQ?ref=fed_asin_title&th=1)	1	\$77.94
Power Supply Adapter	S-60-5	Power Supply Adapter, 5V, 12A, 60W	Chiffonade (https://www.amazon.com/dp/B0FR42VBWX?ref=fed_asin_title)	1	\$12.99
¾ Inch Wood	PLY-12-24-34	Wood used for the enclosure	Woodpeckers (https://www.amazon.com/dp/B0GL9L9RLQ?ref=fed_asin_title&th=1)	1	\$59.99
¾ Inch Wood	B0BX4RF9BF	Wood used for the back wall of the enclosure	MakerStock (https://www.amazon.com/dp/B0BX4RF9BF?ref=fed_asin_title&th=1)	1	\$38.94
Acrylic Plexiglass	B088BPVVR3	Acrylic used for the side and front wall of the enclosure	CALPALMY (https://www.amazon.com/dp/B088BPVVR3?ref=fed_asin_title&th=1)	1	\$66.49
PCB #1	N/A	PCB of the electronics	N/A	1	\$108.68
Arduino UNO REV3	ARD_A000066	Arduino used to run the code	Arduino (https://www.amazon.com/dp/B008GRTSV6?ref=fed_asin_title&th=1)	1	\$27.05
PCB #2	N/A	PCB of the electronics	N/A	1	\$61.75

Costs: \$723.91

Figure A6: Bill of Materials and Final Cost

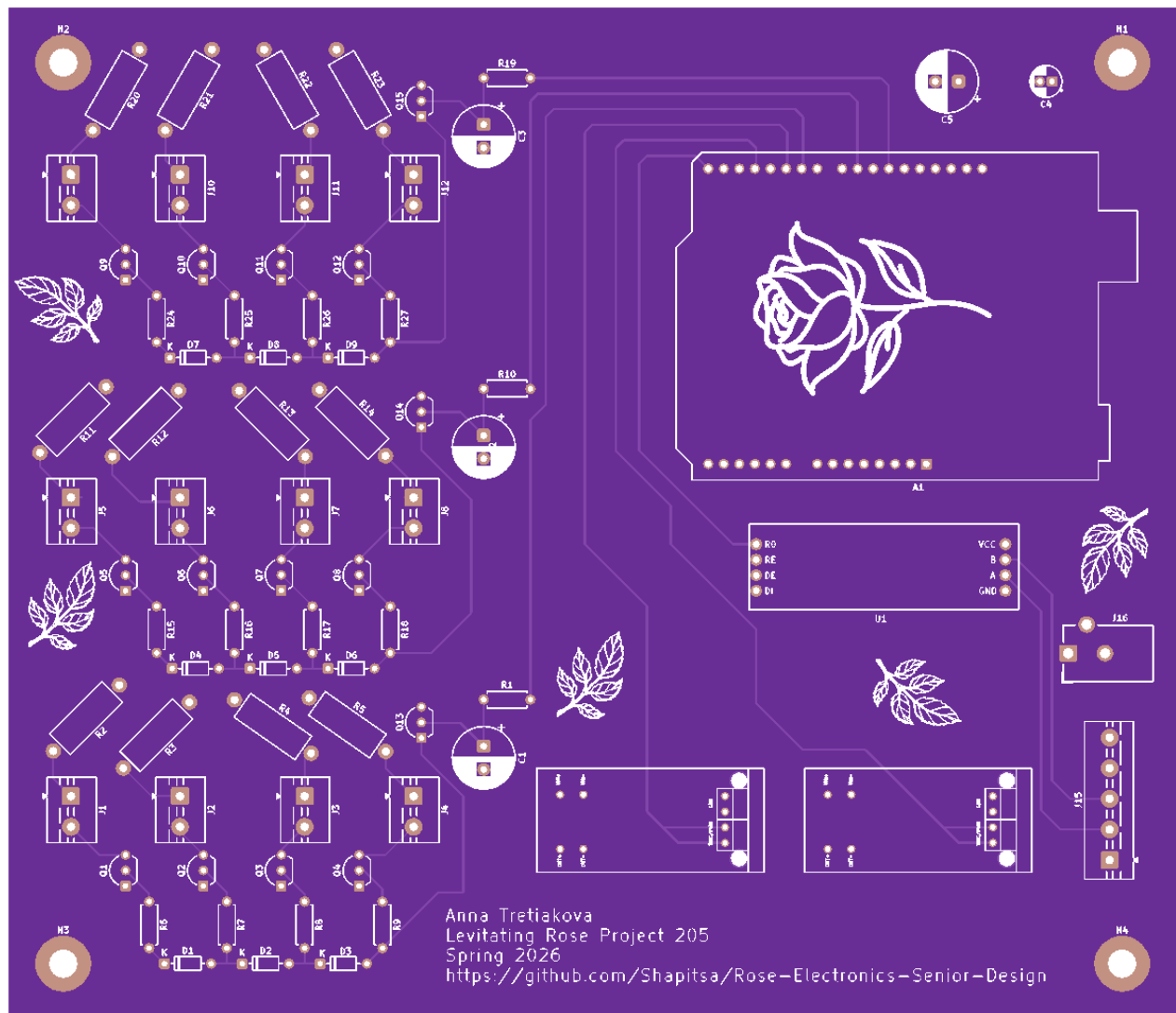


Figure A7: Complete PCB

```

#include <DMXSerial.h>
const int outputPin1 = 6;
const int outputPin2 = 9;
const int outputPin3 = 11;
const int outputPin4 = 5;
const int outputPin5 = 3;
const int startChannel = 0 * 5 + 1;
#define pin1DefaultLevel 0
#define pin2DefaultLevel 0
#define pin3DefaultLevel 0
#define pin4DefaultLevel 0
#define pin5DefaultLevel 0

void setup() {
    DMXSerial.init(DMXReceiver);
    pinMode(outputPin1, OUTPUT);
    pinMode(outputPin2, OUTPUT);
    pinMode(outputPin3, OUTPUT);
    pinMode(outputPin4, OUTPUT);
    pinMode(outputPin5, OUTPUT);
}

void loop() {
    unsigned long lastPacket = DMXSerial.noDataSince();
    if (lastPacket < 5000) {
        int ch1 = DMXSerial.read(startChannel);
        int ch2 = DMXSerial.read(startChannel + 1);
        int ch3 = DMXSerial.read(startChannel + 2);
        int ch4 = DMXSerial.read(startChannel + 3);
        int ch5 = DMXSerial.read(startChannel + 4);
        analogWrite(outputPin1, ch1);
        analogWrite(outputPin2, ch2);
        analogWrite(outputPin3, ch3);
        analogWrite(outputPin4, min(ch4, 140));
        analogWrite(outputPin5, min(ch5, 220));
    }
}

```

Figure A8: Complete Arduino Code

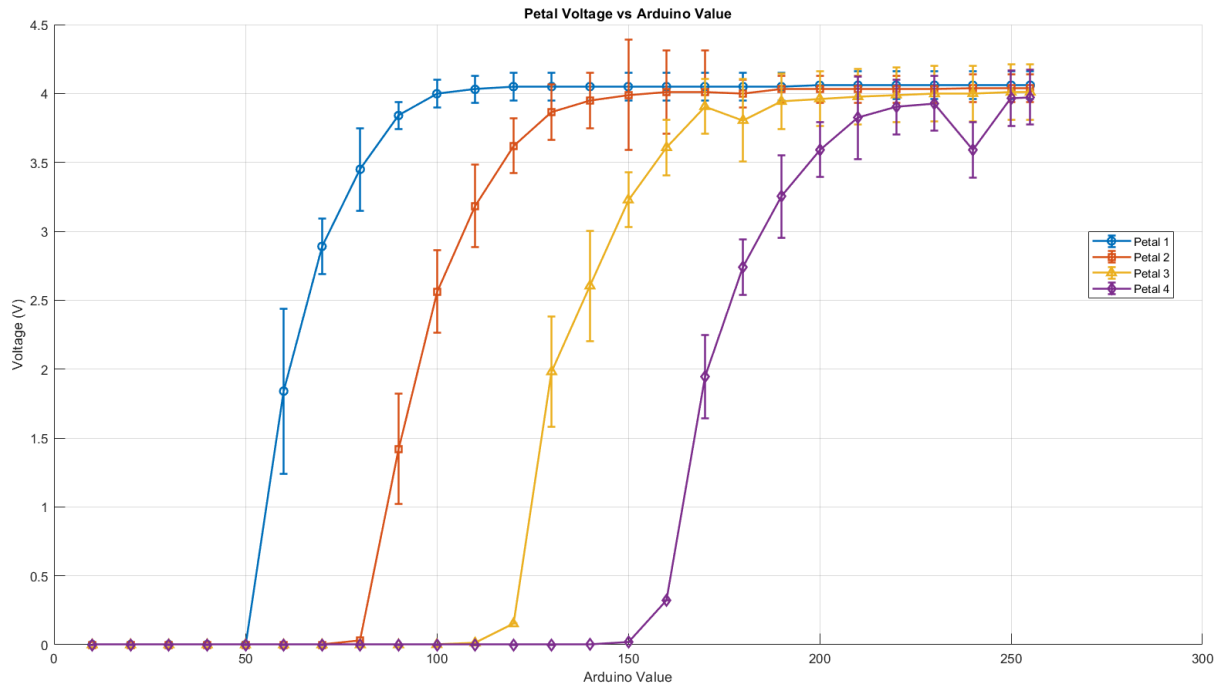
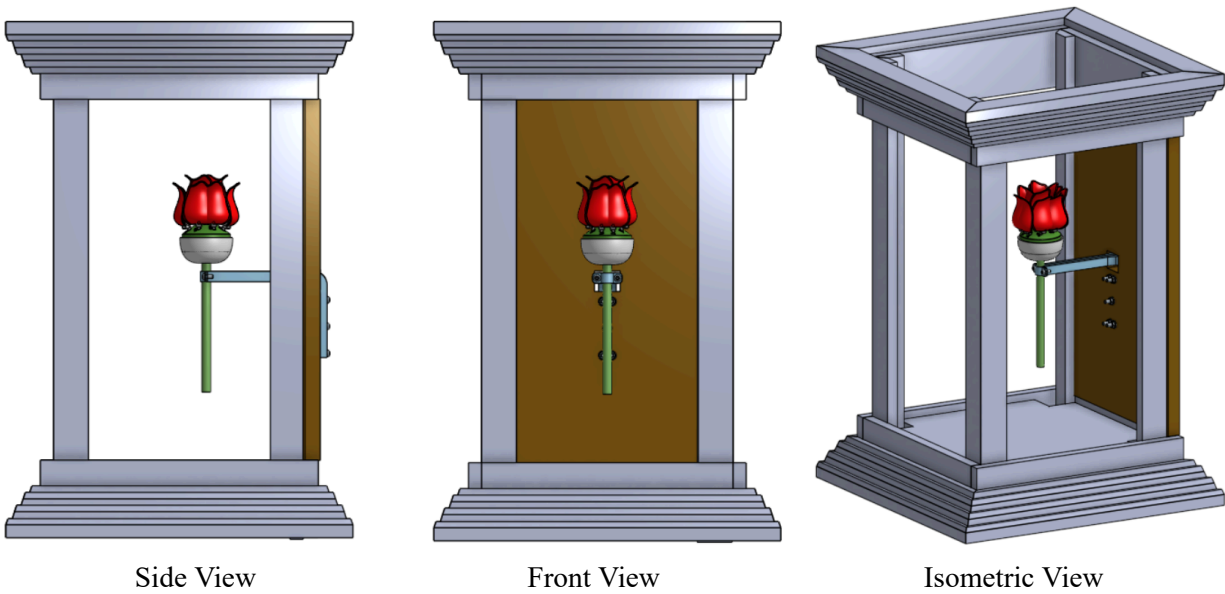
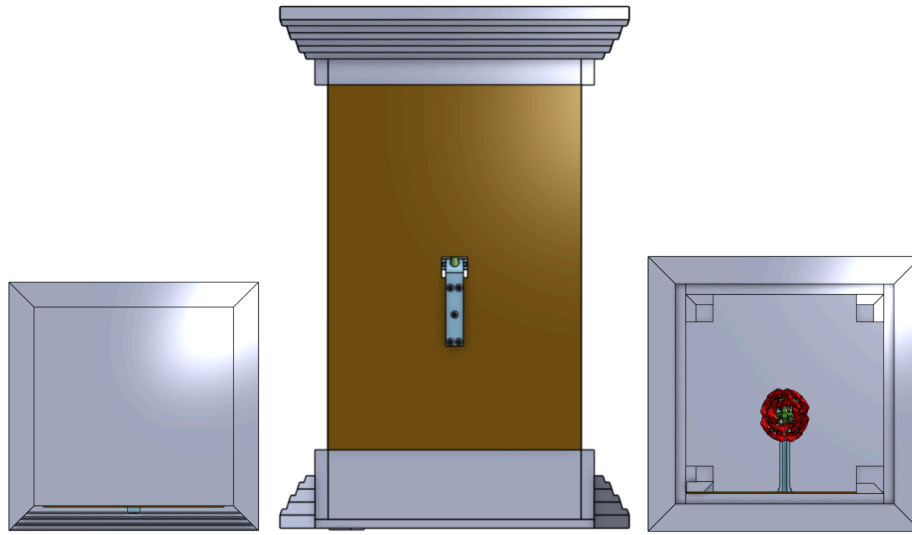


Figure A9: Preliminary Prototype Testing - Petal Voltage vs Arduino Value

Figure A10: Preliminary Prototype Testing - LED and UV Light Voltage vs Arduino Value





Bottom View

Back View

Top View (Roof Transparent)

Figure A11: Full CAD Assembly of Enclosure